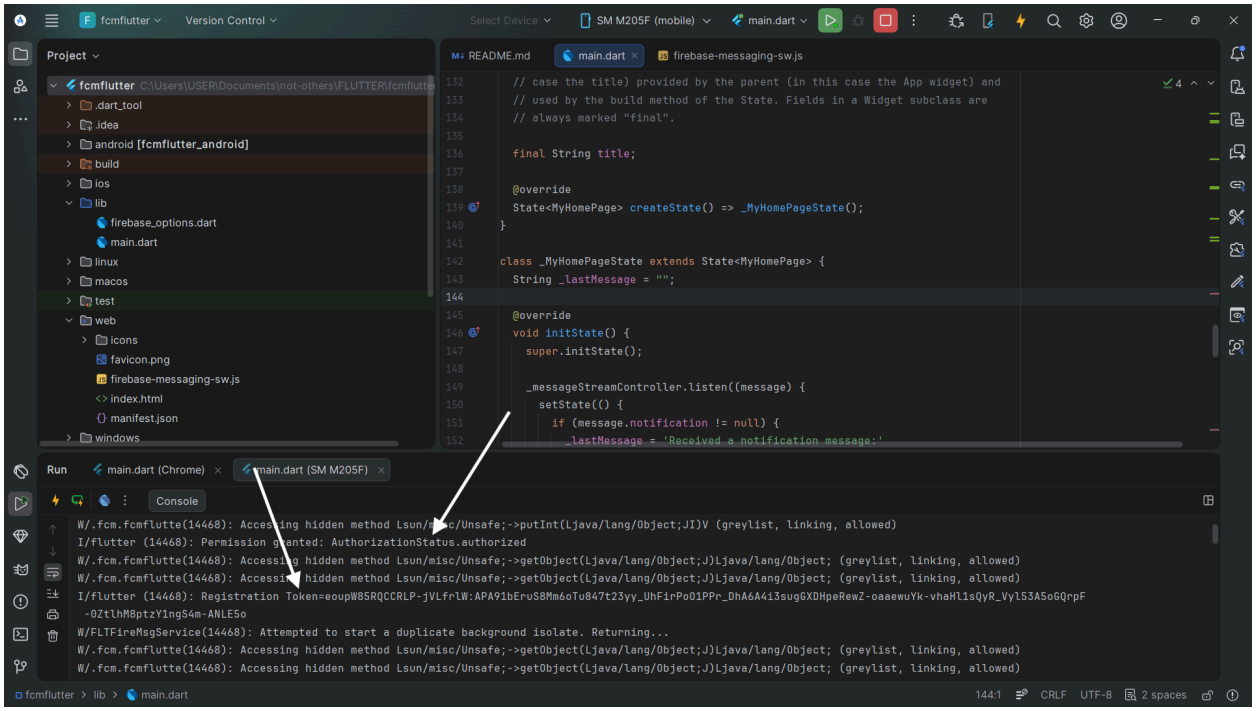
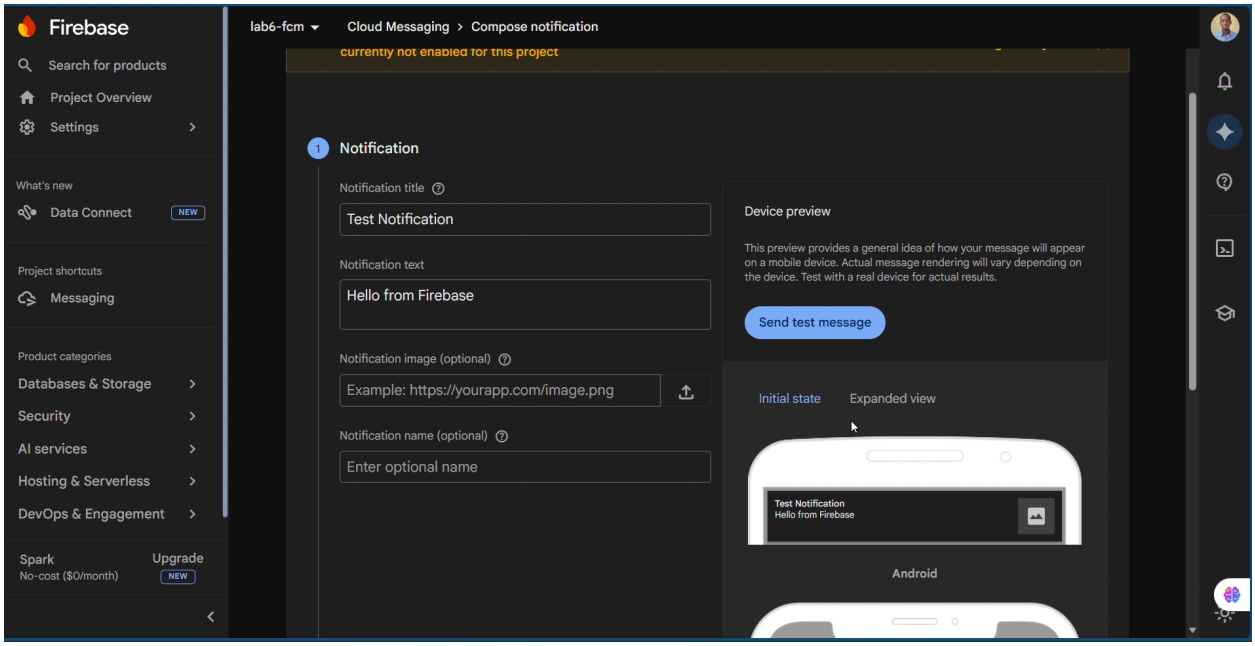
Lab Report: Push Notifications with Firebase Cloud Messaging

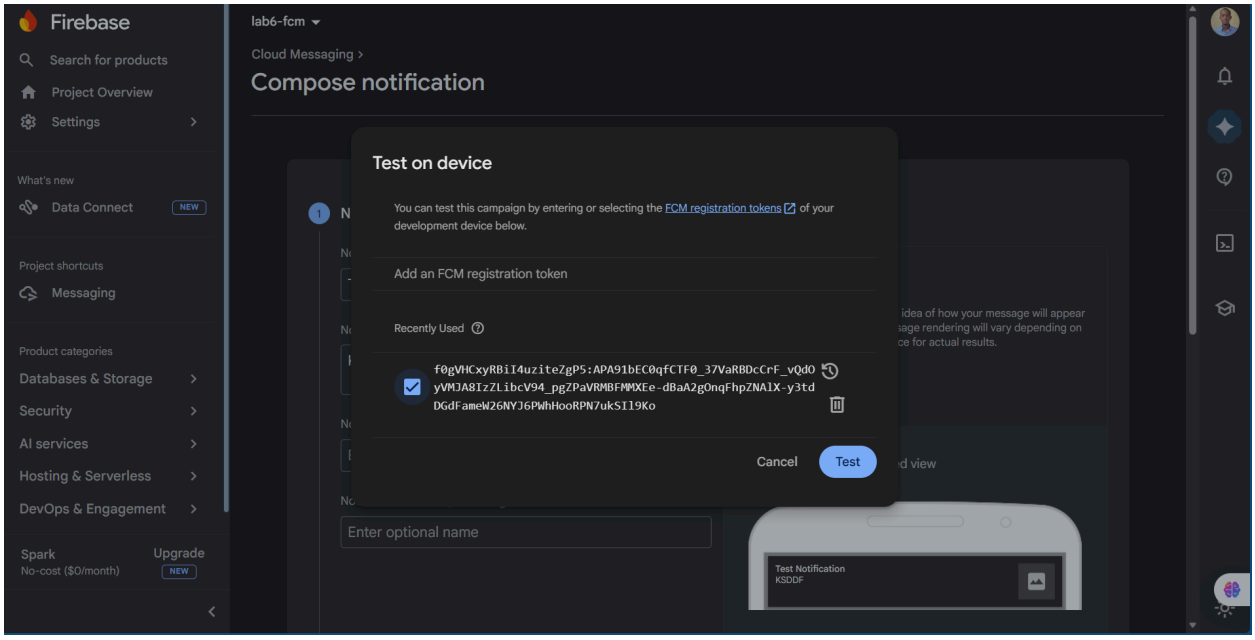
1. Screenshots



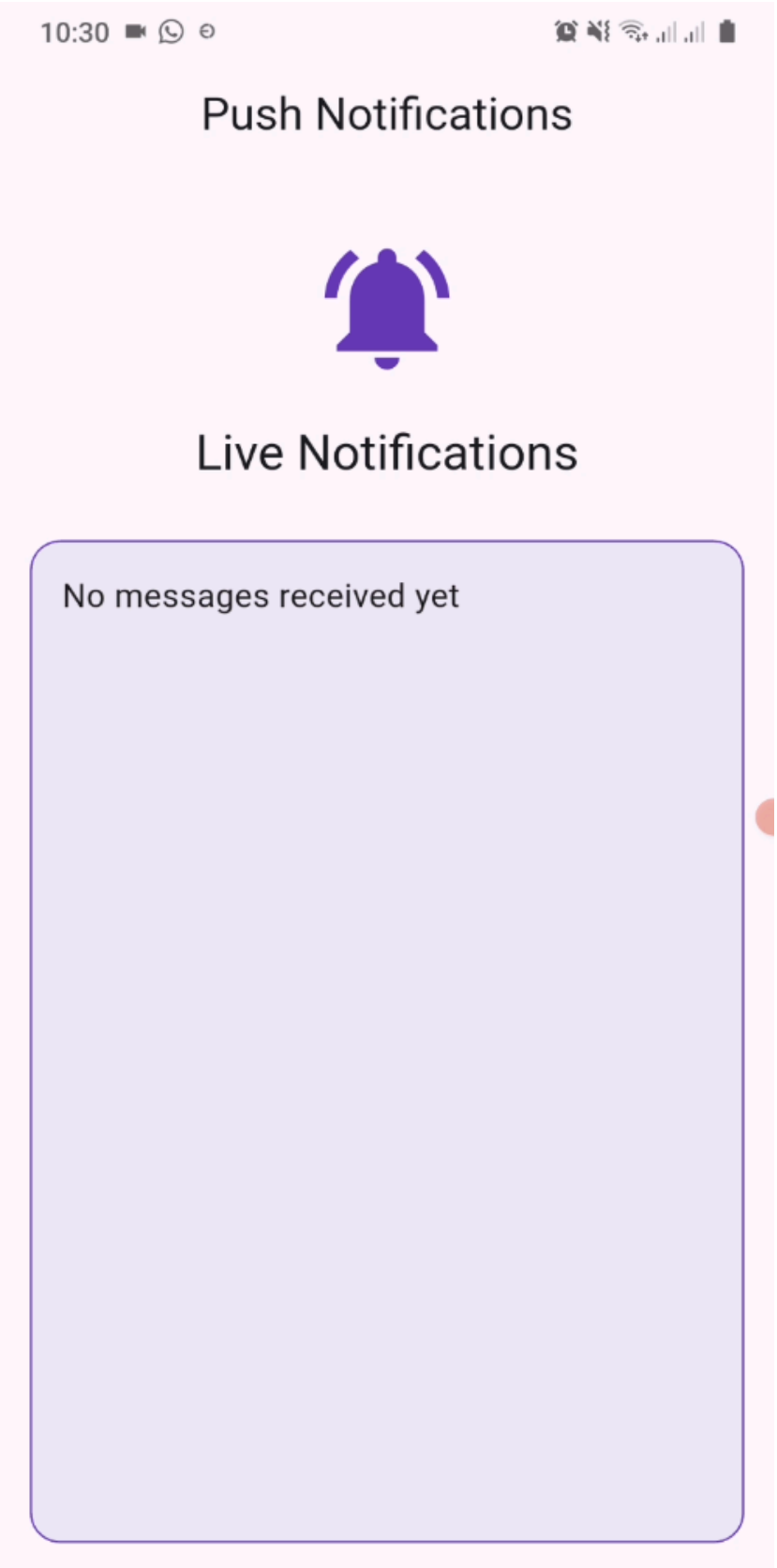
Screenshot - Notification Permission granted and the Registration Token given



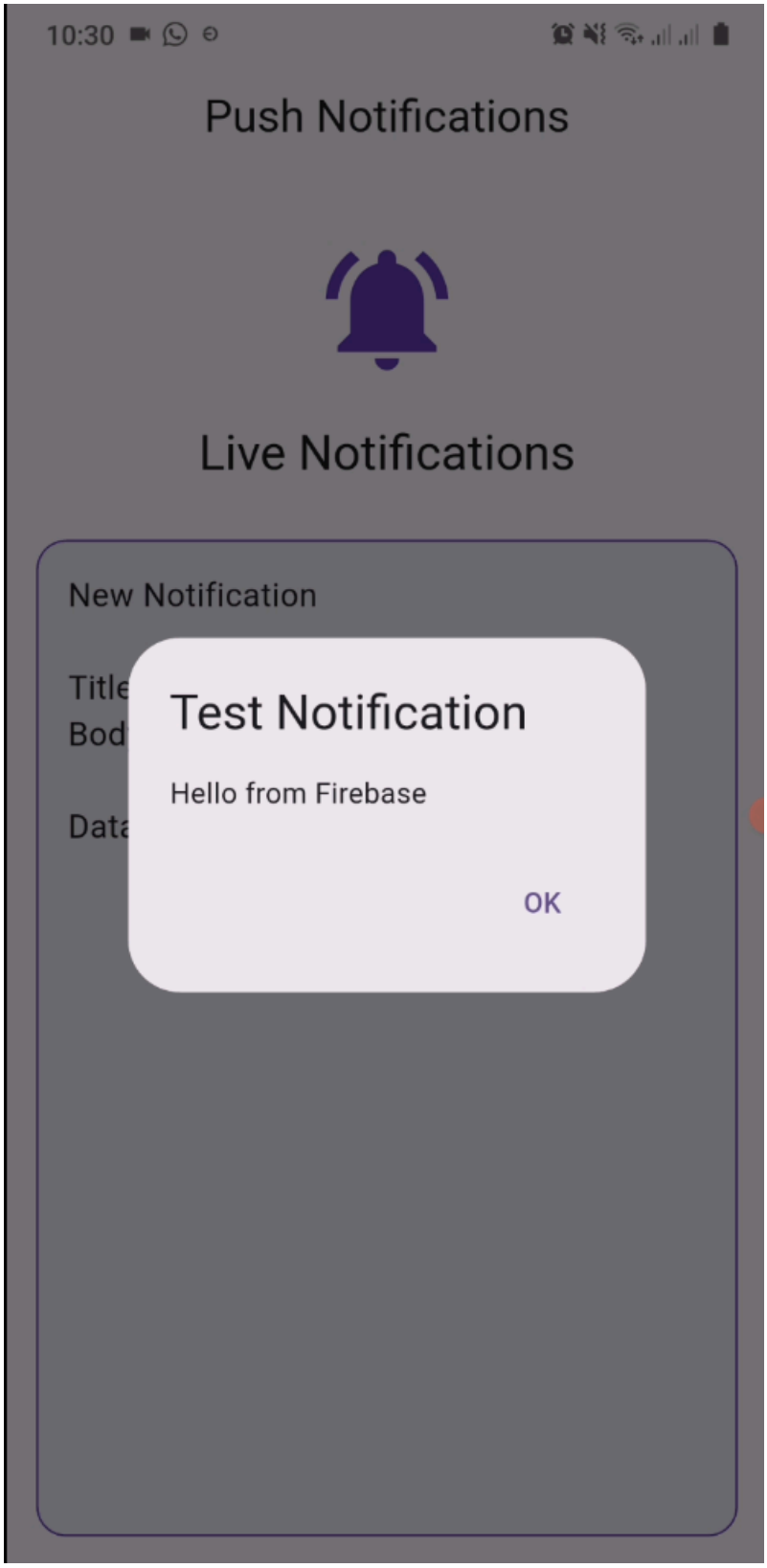
Screenshot - Creating a notification in Firebase Console



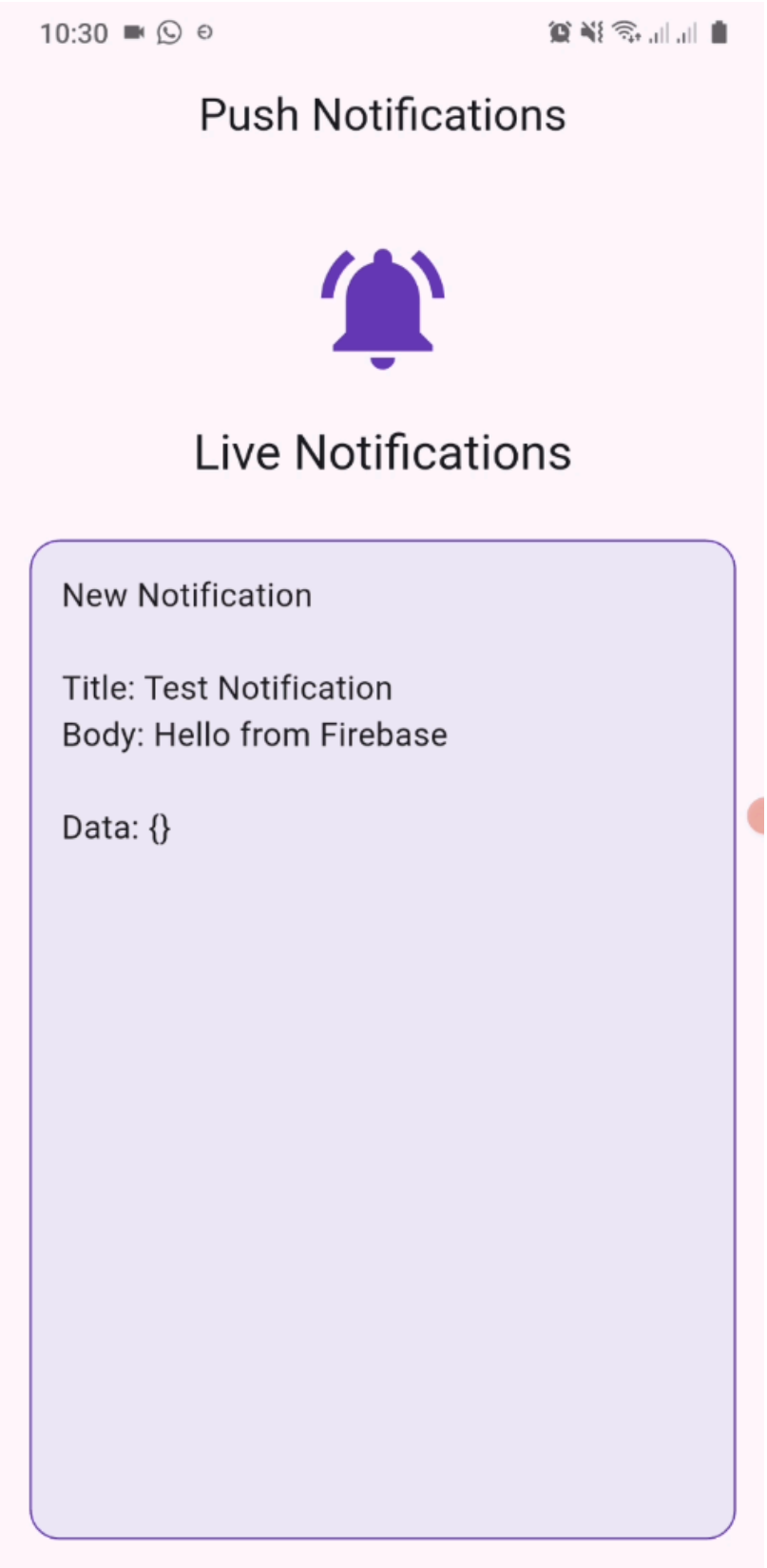
Screenshot - Adding the registration token I copied from the console



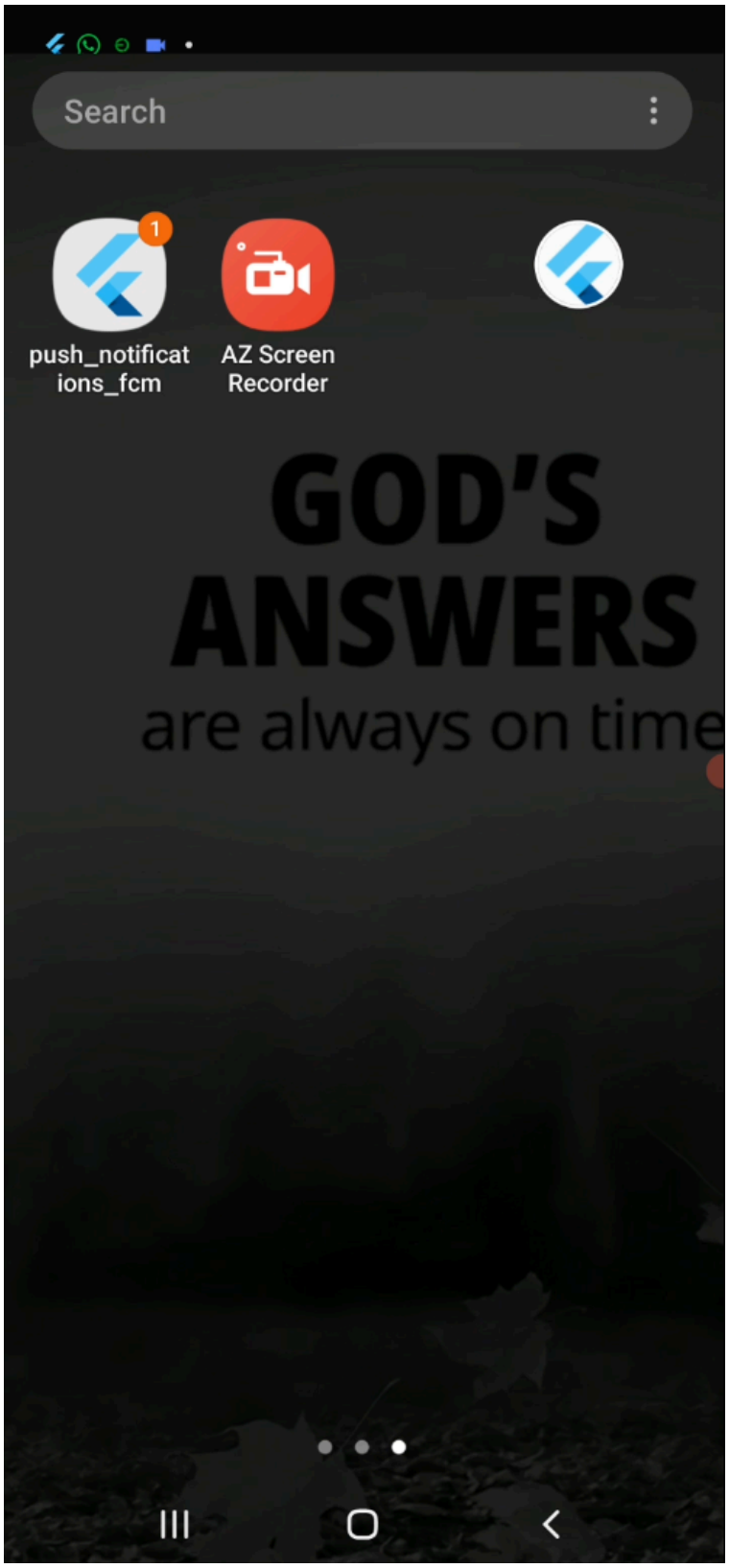
Screenshot - The app UI with no notifications



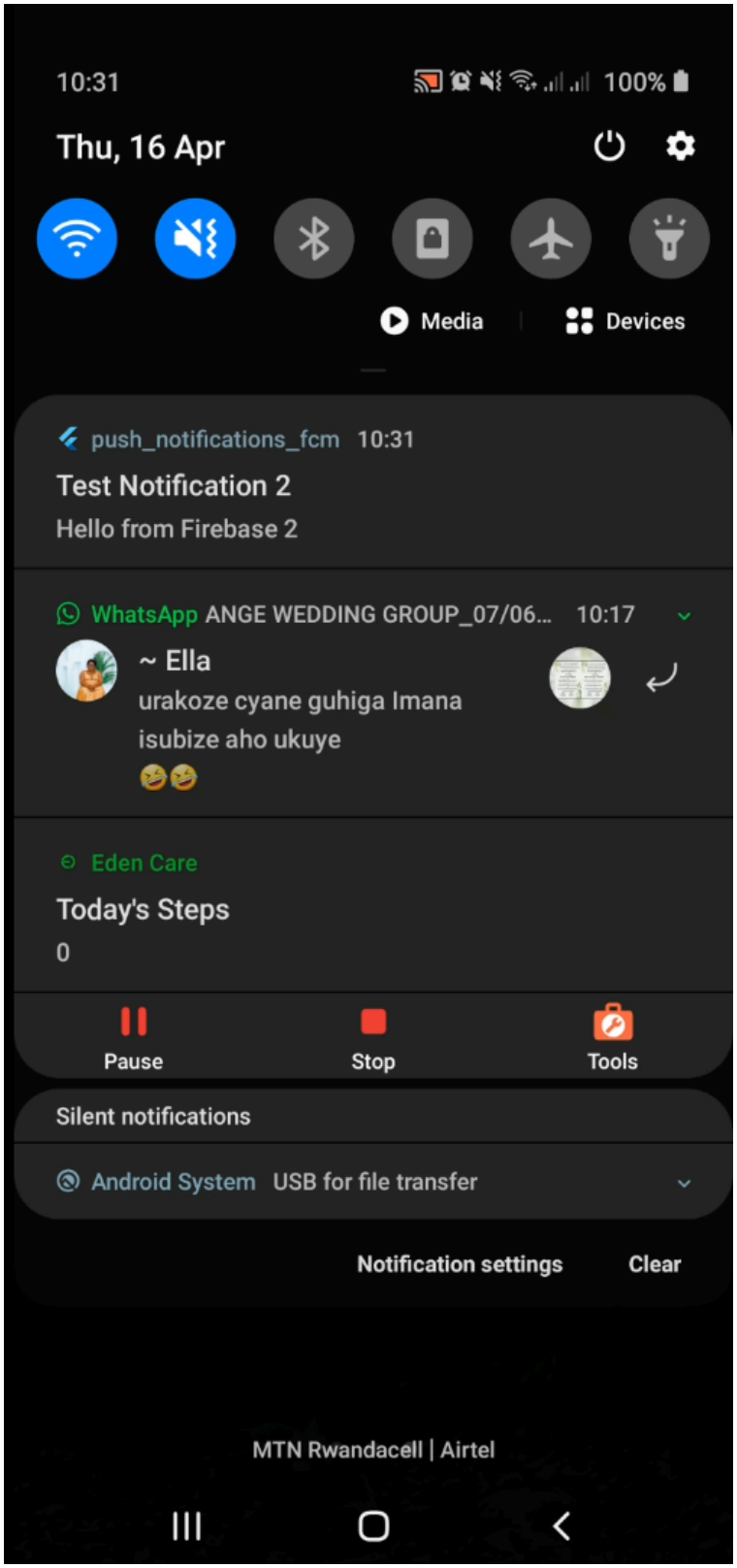
Screenshot - The app receives a new notification and immediately a modal of the message appears



Screenshot - After clicking ok on the modal now the message appear in the app



Screenshot - When the notification comes while the app is not open, a badge appear on the app icon



Screenshot - Also when the notification comes while the app is not open, a notification is in notification panel

2. Steps and Observations

Key steps I followed to make it work:

1. I created a new Flutter project using the command flutter create
2. I created a Firebase project from the Firebase Console and registered my application
3. I installed dependencies:
   * Firebase_Core
   * Firebase_messaging
4. I configured Firbase in my Flutter app using the FlutterFire CLI by running flutterfire configure
5. I initialized Firebase in the main.dart file using Firbase.initializeApp()
6. I requested notification permissions from the user using the requestPermission() method
7. I registered the application with Firebase Cloud messaging and retrieved the device registration token using getToken().
8. I printed the device token in the Console and saved it for testing notifications.
9. I implemented a foreground message handler using FirebaseMessaging.onMessage.listen() to receive notifications when the app is not active
10. I implemented a background message handler using FirebaseMessaging.onBackgroundMessage() to receive notifications when the app is not active.

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11. I displayed the received messages inside the app UI by updating the state whenever a new message is received.

12. I sent a test notification using firebase Console by pasting the device token and writing a message.

13. I ran the application on a real device and confirmed that notifications were received successfully in both foreground and background

Observations

- Firebase Cloud Messaging allows real-time communication ~~between~~ between the server and the mobile application.
- Each device receives a unique registration token which is required to send notifications specifically to that device.
- The app must request permission from the user before receiving notifications.
- Notifications behave differently depending on the app state:
  * In the foreground, messages must be handled manually and displayed in the UI.
  * In the background, notifications appear automatically in the notification bar.
- Firebase initialization is required before using any firebase service.

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- The integration process requires correct configuration using FlutterFire CLI and Firebase console.
- The device token may change over time, so it must be updated when sending notifications.
- Handling background messages requires a separate function outside the main application logic.
- Push notifications can include both notification content (title and body) and data payload.
- Testing on a real device is important because some notification features may not work properly on emulators.

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3. Video Demonstration (Extra)

In place of a long series of static images, a video was recorded to demonstrate the real-time functionality of the notification system.

Video Link

What is demonstrated in the video:

1. **Token Retrieval:** Showing the device token being printed/displayed.
2. **Background Test:** Sending a message from Firebase and seeing it arrive in the notification bar while the app is minimized.
3. **Foreground Test:** Sending a message while the app is open and seeing the required popup and UI update.

Reference:

- Firebase